The fNIRS Foundation Model Audit: Independent Evaluation of MtReNIRS and the Case of the Random Encoder

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Foundation models have recently entered functional near-infrared spectroscopy (fNIRS). The most prominent of these, MtReNIRS, was released in October 2026 with reported cross-subject transfer gains on several public datasets. To our knowledge, however, no released fNIRS foundation model has yet been evaluated by a party other than its authors. We report an independent audit of MtReNIRS on a held-out dataset that is absent from its twelve-dataset pretraining corpus (17 participants, 1,020 epochs, three auditory conditions). Using strict leave-one-subject-out evaluation with permutation nulls, a frozen pretrained encoder reaches 0.469 ± 0.095 three-class accuracy (chance 0.333). A randomly initialised encoder of the same architecture, averaged over ten seeds, reaches 0.448 ± 0.010; the pretraining advantage is +2.04 percentage points and is not statistically significant ($p = 0.19$, 95 % CI $[-0.8, +4.9]$ pp). The architecture does significantly outperform hand-crafted features (+5.88 pp, $p = 0.014$). Short-separation regression, currently recommended for fNIRS decoding, reduced accuracy by 4.5 pp and eliminated the architecture advantage, indicating that part of the decodable signal is systemic. Naive feature stacking of encoder and hand-crafted representations significantly degraded performance (−4.71 pp, $p = 0.042$). We argue that *randomly initialised encoders of the same architecture* be adopted as a mandatory baseline in fNIRS foundation-model evaluation, and we release our harness together with a companion detectability calculator for study design.

1 Introduction

Functional near-infrared spectroscopy (fNIRS) measures cortical haemodynamics through the scalp at a fraction of the cost and immobility of fMRI, and has become a practical modality for brain–computer interfaces and neuroergonomics. Its decoding performance, however, has historically been limited: sparse montages, superficial physiological contamination and modest spatial specificity constrain single-trial accuracy [7].

Foundation models are now being proposed as the remedy. In late 2026, two fNIRS foundation models appeared within weeks of each other—MtReNIRS [6] and GeoNIRS [3]—alongside a curated 67-dataset corpus of 1,955 hours [4]. MtReNIRS reports leave-one-subject-out gains over a from-scratch copy of the same network, and over feature-engineered baselines, on several public datasets.

These are promising results, but they share a structural weakness: every published number comes from the model's own authors, on datasets most of which are part of the model's pretraining corpus. Other neuroimaging modalities have already confronted this. EEG now has independent

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benchmark efforts spanning dozens of models and tasks; fNIRS has no equivalent, and we are not aware of any published third-party evaluation of a released fNIRS foundation model. This matters because recent large-scale work in the neighbouring EEG literature has found that pretraining advantages can be small, inconsistent across tasks, or absent relative to compact convolutional baselines.

This paper reports such an evaluation. We audit MtReNIRS on a dataset that is verifiably absent from its pretraining corpus, using a protocol fixed in advance: strict leave-one-subject-out cross-validation, identical folds for every arm, matched feature dimensionality, permutation nulls, and paired statistical tests across folds. Our central question is deliberately blunt: *does the pretrained encoder beat a randomly initialised encoder of the same architecture?* The random-encoder control is routine in vision and language, where it separates the contribution of pretraining from the contribution of architecture; in fNIRS foundation-model papers it is, to our knowledge, not used at all.

Our contributions are:

1. An independent, contamination-aware evaluation of a released fNIRS foundation model on a clean held-out dataset, with permutation-calibrated significance and paired tests.

2. A ten-seed random-initialisation control, which shows that the pretraining advantage is +2.0 pp and not statistically significant, while the architecture's advantage over hand-crafted features is significant.

3. A short-separation ablation showing that the currently recommended physiological regression *reduces* decoding accuracy by roughly 4.5 pp on this paradigm and eliminates the architecture advantage—evidence that part of the decodable signal is systemic rather than cortical.

4. A negative result for naive feature-bank stacking, and a statistical demonstration that the feature combination must be learned rather than concatenated.

5. A companion detectability calculator, derived while diagnosing a trigger-free in-house dataset, which quantifies the smallest evoked response a given montage and design could detect. We include it because it is the tool we used to establish that our own data could not support the analysis we had planned—a check we believe should precede decoding claims in this literature.

We emphasise at the outset what this paper does *not* claim. It is not evidence that fNIRS foundation models are useless, nor that pretraining cannot help. It is a single-dataset, single-model audit with limited statistical power, and its result is a failure to establish an advantage, not a demonstration of its absence.

2 Background

2.1 fNIRS decoding and its constraints

fNIRS records changes in oxygenated (HbO) and deoxygenated (HbR) haemoglobin concentration from near-infrared light passing between sources and detectors on the scalp. The signal is haemodynamic, slow (peak responses 5–7 s after onset), and contaminated by superficial physiology—cardiac pulsation, respiration, and Mayer waves—which is measured directly by *short-separation channels*, i.e. source–detector pairs with separations of roughly 8–15 mm that sample scalp and skull rather than cortex.

Two constraints dominate decoding. First, spatial resolution: a recent analysis of the resolution required for image reconstruction reports that at 2 cm—the continuous-wave fNIRS regime— 300-way retrieval falls to about 20 %, against 93 % for full-resolution fMRI [1]. Second, probe density: single-trial decoding accuracy improves systematically with optode density [7]. Sparse montages such as the one we audit here (33 source–detector pairs) sit at the low end of that curve.

2.2 Foundation models across neuroimaging

Pretrained representation models now exist for several modalities. For EEG: LaBraM [5], CBraMod [13], EEGPT [12], and unified EEG/MEG models such as BrainOmni [2]. For fNIRS, MtRe-NIRS [6] is a 1.23M-parameter spatial–temporal transformer pretrained on twelve public datasets (1,387 runs, 203,420 ten-second windows, exceeding 567 hours) using one 32 GB V100, producing a 128-dimensional embedding per channel and 10-second window and accepting arbitrary montages through standard 10-05 channel coordinates. GeoNIRS [3] takes a geometry-aware approach, and a curated 67-dataset corpus [4] now supports further pretraining.

A parallel line of work has developed architectures tailored to fNIRS physiology and montage geometry rather than adapting generic backbones: transformer and general vision frameworks for fNIRS classification [15, 16], representations that model the delayed haemodynamic response explicitly [17], a unified framework informed by local brain activation patterns [14], and a domain-knowledge-fused state-space model [18]. That work establishes that architecture and physiological inductive bias matter for fNIRS decoding. It is also the premise our random-encoder control tests: if architectural inductive bias is what carries the performance, a randomly initialised encoder of the same architecture should recover much of it—and, in our results, it does.

2.3 The evaluation gap

Evaluation practice has not kept pace. For EEG, independent benchmarks now evaluate dozens of models across many tasks and datasets. For fNIRS we could find no independent benchmark: the published evaluations of both released models are by their own authors, and much of the evaluation data overlaps the pretraining corpora, which makes apparent transfer gains difficult to attribute. In the EEG literature the analogous concern is already empirically grounded: large-scale pretraining has been reported to yield no consistent advantage over compact convolutional baselines on some downstream tasks, and the benefit of pretraining has been found to depend on corpus scale and on the specific benchmark.

The control we adopt here is standard elsewhere but absent from fNIRS foundation-model papers. In vision and language, a randomly initialised model of the same architecture is the accepted baseline that separates *architecture* from *pretraining*; the idea has an established lineage in the random-features literature [9], where fixed random projections are known to be surprisingly competitive for many tasks. If a pretrained encoder cannot beat a random one, the claim that pretraining is what produces the performance is unsupported regardless of how good the absolute accuracy is.

2.4 Preprocessing as a confound

Physiological noise regression is widely recommended: Fischer et al. [7] integrate short-separation regression into cross-validation and report gains of roughly 4 % within-subject and more than 10 percentage points in cross-dataset transfer, and argue that image-space (parcellated) representations further improve generalisation. Yet the same literature warns that decoding results can be

inflated by non-neural structure: Rybář et al. [10] show that cue presentation windows grossly overestimate semantic BCI performance. Our short-separation ablation (Section 4.2) speaks directly to this tension.

3 Methods

3.1 Dataset

We use the passive auditory fNIRS dataset of Luke et al. [8], distributed as BIDS with SNIRF recordings. It contains 17 participants, each viewing a block-design protocol with three conditions (Control, Speech, Noise), 20 events per condition per participant (60 events each), 5-second stimulus duration and pseudo-randomised inter-stimulus intervals. Raw data are continuous-wave amplitudes at two wavelengths (760 and 850 nm), recorded at 5.2083 Hz with 66 measurement channels over 12 sources and 20 detectors.

The montage is identical across participants: 33 source–detector pairs, of which **8 are short-separation channels** (7.6–8.4 mm) and **25 are long channels** (30.4–41.9 mm). This is important for Section 4.2.

Why this dataset. MtReNIRS's twelve pretraining datasets are published in its repository as aligned-montage summaries: Ajra2026, Chen2023, Defenderfer2019, Ferron2019, Gao2023, Hudak2017, Li2020, Morgan2026, Shin2017, Spapé2023 (NEMO [11]), Sudan2026-cocktail-visual, and Xiafei2024-driving. The present dataset is not among them, so it is a clean held-out test set. By contrast NEMO—the most attractive public image-viewing fNIRS dataset—*is* in the pretraining corpus and is therefore unsuitable as a headline test.

3.2 Preprocessing

Raw intensities were converted to optical density and then to HbO/HbR concentration changes using the modified Beer–Lambert law with tabulated extinction coefficients, a differential pathlength factor of 6.0, and channel-wise source–detector distances computed from the three-dimensional optode coordinates. Signals were linearly detrended and band-pass filtered at 0.01–0.2 Hz (third-order Butterworth, zero-phase). Each event contributed a 10-second epoch starting at stimulus onset, resampled to 10 Hz (100 samples, the MtReNIRS window contract) and z-scored per epoch and channel.

Channel positions for the model were obtained by averaging each source–detector pair and projecting the midpoints to standard 10-05 coordinates using the model's alignment toolkit (MtSi1005); all 66 measurement channels mapped to valid slots, with a median projection distance of 13.2 mm.

3.3 Model and arms

MtReNIRS was loaded from its bundled MIT-licensed checkpoint (`mtrenirs_10s.pt`, 1,226,368 encoder parameters) and run in inference mode on CPU. We compare four arms under one protocol:

Pretrained frozen MtReNIRS embeddings, flatten-pooled (66 channels × 128 dimensions).

Random-init the same architecture with all encoder parameters re-initialised to their default initialisation, repeated over **ten seeds**; each seed produces one arm.

Classical per-channel mean and linear slope over the epoch (132 features).

Stack pretrained and classical features concatenated after block-wise PCA.

3.4 Probe and statistics

Every arm passes through an identical head. For each of the 17 leave-one-subject-out folds we standardise features on the training participants only, reduce dimensionality with PCA fitted on training data only, re-standardise, and fit multinomial logistic regression ($C = 1$, 3,000 iterations; no fold failed to converge). Total dimensionality is held constant at 64 across all arms, so a two-block stack receives PCA-32 per block and cannot win merely by having more features.

We report mean ± standard deviation of fold accuracy, paired t-tests and Wilcoxon signed-rank tests across folds, and bootstrap 95 % confidence intervals over folds. Significance of the embedding arms is further calibrated against permutation nulls: 100 label permutations, re-using the per-fold PCA (which is unsupervised, so no information from the test labels enters). Chance is 33.3 %.

3.5 Preprocessing ablation

The main experiment (Section 4.1) includes all 33 pairs, i.e. short-separation channels enter the model as if they were cortical channels. To separate the contributions we compare three variants:

- **raw**: all 33 pairs, no regression;
- **long**: the 25 long pairs only, short channels excluded from the model input;
- **wss**: the 25 long pairs after regressing out the eight short-separation signals from every long channel, per participant, on the continuous signal before epoching (ordinary least squares with all eight short channels and an intercept; the fitted short-channel component is subtracted). We verified that this removes the shared variance: the mean absolute correlation between long channels and the mean short-channel signal falls from 0.434 to 0.000.

Regression is performed on continuous data, never per epoch, so no epoch-level information enters the cleaning step.

3.6 Companion detectability calculator

While attempting to analyse a separate trigger-free pilot recording, we built a small tool that we believe is independently useful and include here because it is the basis on which we decided *not* to make decoding claims about that recording. The pilot data are informal, self-collected exploratory recordings made by the author without a formal ethics protocol; they are not distributed, they are referred to only by neutral session codes (A and B), and they contribute no result beyond the illustration in §4.4.

The tool injects synthetic haemodynamic responses of specified amplitude at a candidate design's event onsets into a recording of the same participant's continuous signal, then runs the same onset-recovery or decoding pipeline used on real data. Sweeping injected amplitude yields a *detectability curve*: the probability that a response of a given size would have been detected at the sample size, montage and noise level actually available. This design logic follows the synthetic-HRF validation component of Fischer et al. [7], implemented in Cedalion; our addition is the pre-study power framing and the null calibration described next.

Calibration requires care. A statistic that maximises over candidate onsets cannot be calibrated by circularly shifting the data, because shifting the data is equivalent to shifting the template family: the null maximum then equals the observed maximum and the test has no power. We

Table 1: Main comparison (all 33 pairs, 17 leave-one-subject-out folds, 64 matched dimensions). Chance is 0.333. CIs are bootstrap 95 % intervals over folds for the difference.

Arm	LOSO accuracy	Δ vs pretrained	paired p (95 % CI)
MtReNIRS (pretrained)	$\mathbf{0.4686 \pm 0.0955}$	—	—
Random-init, mean of 10 seeds	0.4482 ± 0.0102	+2.04 pp	0.188 [−0.82, +4.86]
Classical features	0.4098 ± 0.0754	+5.88 pp	0.014 [+1.86, +10.00]

therefore use phase-randomised surrogates, which preserve the power spectrum while destroying the phase relationship to the design, and we report this explicitly because the failure mode is easy to reproduce.

3.7 Reproduction

All code, fixed protocols and result files are released (§7). Preprocessing, alignment, evaluation and figure generation are separate scripts; every number in this paper can be regenerated from the raw BIDS dataset.

4 Results

4.1 Does the pretrained encoder beat a random one?

Table 1 reports the main comparison on the full 66-channel input. Both embedding arms are far above the permutation null (null mean 0.332, 95th percentile 0.364; $p < 0.01$ for both), so the task is decodable and the pipeline is sound.

The pretrained encoder is numerically the best arm, but **its advantage over a randomly initialised encoder of the same architecture is +2.04 pp and not statistically significant** ($p = 0.188$; Wilcoxon $p = 0.225$; 95 % CI $[-0.82, +4.86]$ pp). The ten random seeds are tightly clustered (0.4482 ± 0.0102, range 0.4353–0.4696), so this is not seed noise; the pretrained model sits at the top of that distribution but the sample of 17 folds cannot resolve the difference.

By contrast, the architecture does significantly outperform hand-crafted features (+5.88 pp, $p = 0.014$), and the random encoder alone also beats classical features (+3.84 pp, $p = 0.039$). The reading we take is that *the spatial–temporal transformer representation*, not the pretraining, supplies most of the measurable benefit on this task.

4.2 Short-separation regression reduces decoding accuracy

Table 2 and Figure 1 report the preprocessing ablation. Dropping the eight short-separation channels from the model input costs 1.86 pp (pretrained arm, $p = 0.318$), and additionally regressing their contribution out of the long channels costs a further 2.65 pp, for a total of 4.51 pp between *raw* and *wss* ($p = 0.102$). Individually these differences are not significant at 17 folds, but the ordering is consistent in every arm and the pattern is monotone.

Two features of this ablation matter. First, the *architecture* advantage over classical features is significant before cleaning ($p = 0.014$ and $p = 0.006$) and disappears after cleaning ($p = 0.297$). Second, the accuracy of the classical arm is essentially unaffected by the same manipulation ($0.4098 \rightarrow 0.3931 \rightarrow 0.4049$). The manipulation therefore does not simply remove noise from all arms equally: it preferentially removes information that the embedding arms were using.

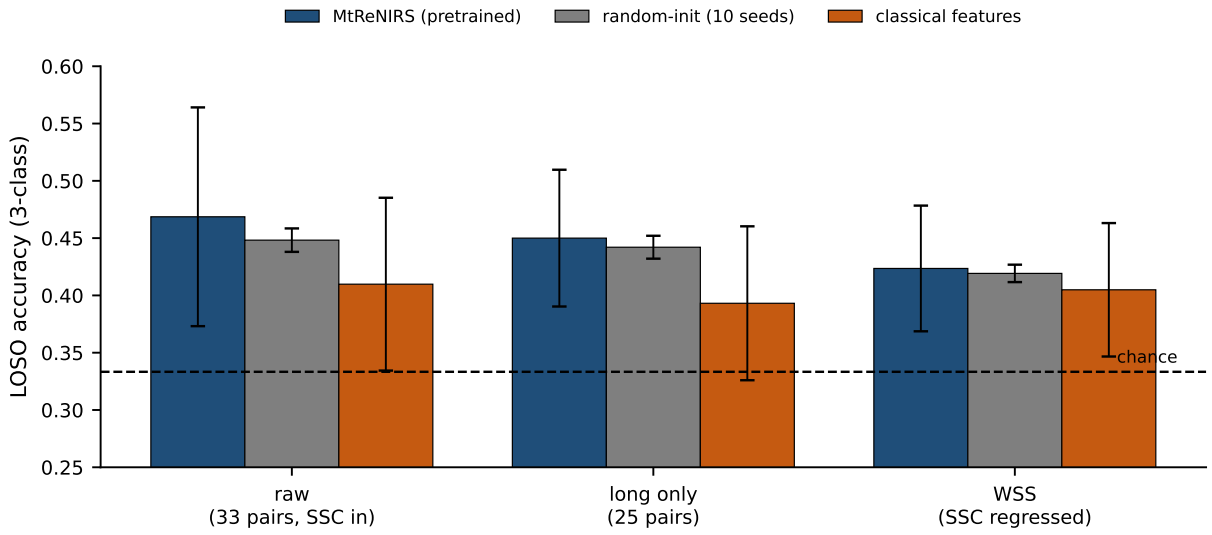


Figure 1: Leave-one-subject-out accuracy by arm and preprocessing variant. Error bars are standard deviations over folds; the dashed line is chance. Removing short-separation channels and, more strongly, regressing them out reduces accuracy and compresses the differences between arms.

Table 2: Preprocessing ablation. WSS = short-separation channels regressed out of long channels before epoching. Each column is the same ten-seed and classical comparison as Table 1.

Variant	Pretrained	Random (10 seeds)	Classical	Pre. − rand.	Pre. − class.
raw (33 pairs)	**0.4686**	0.4482	0.4098	+2.04 (p=0.19)	+5.88 (p=0.014)
long only (25)	0.4500	0.4421	0.3931	+0.79 (p=0.41)	+5.69 (p=0.006)
wss (25, regressed)	0.4235	0.4192	0.4049	+0.43 (p=0.71)	+1.86 (p=0.30)

We interpret this as evidence that a component of the decodable signal in this paradigm is *systemic* rather than cortical, consistent with the warning of Rybář et al. [10] that non-neural structure can inflate decoding. If that reading is correct, short-separation regression does what it is designed to do—and in doing so removes signal, not only noise, when the task is accompanied by global physiological changes that differ between conditions. We treat this as a cautionary result rather than a refutation of the method: it is one paradigm, one dataset, and the contrasts are individually below significance.

4.3 Naive feature stacking significantly degrades performance

At matched total dimensionality, concatenating encoder and classical features is *worse* than the encoder alone (Table 3): -4.71 pp, $p = 0.042$, 95 % CI $[-8.73, -0.59]$. The stack is statistically indistinguishable from classical features alone ($+1.18$ pp, $p = 0.567$), and the same pattern holds for the random-init arm (-2.75 pp, $p = 0.023$).

The per-channel mean and slope features are therefore largely redundant with the transformer embeddings: forcing half the dimensions to carry them costs more information than they add. Any integration of multiple representations for fNIRS should be *learned*—attention or gating over feature banks, or joint training with a shared trunk—rather than concatenated.

Table 3: Feature-bank stacking at matched dimensionality (raw variant).

Arm	LOSO accuracy	Comparison
Pretrained (PCA-64)	$\mathbf{0.4686 \pm 0.0955}$	—
Stack: pretrained + classical (32+32)	0.4216 ± 0.0626	-4.71 pp vs pre., $p = 0.042$
Classical alone	0.4098 ± 0.0754	stack -1.18 pp, $p = 0.567$

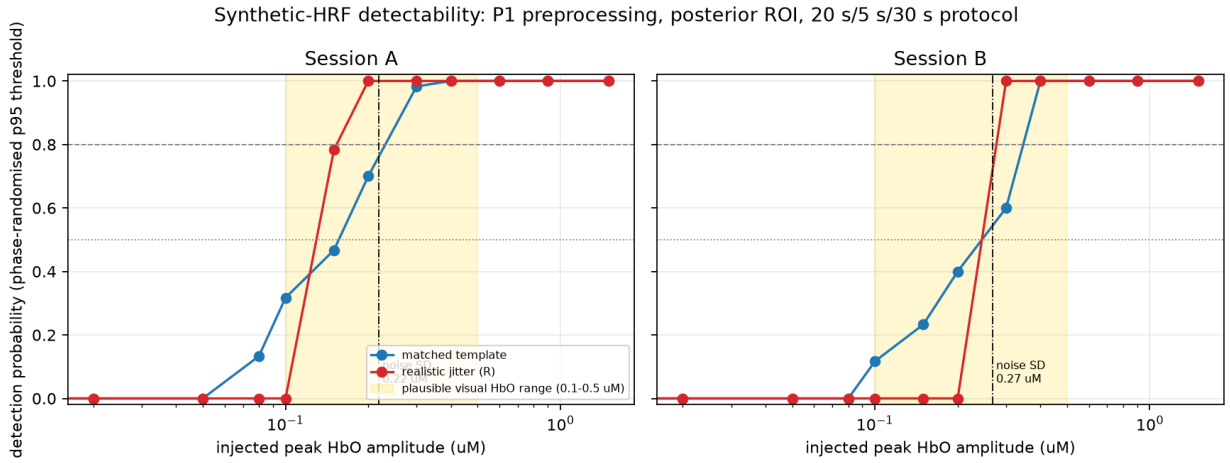


Figure 2: Detectability curves for the two participants of a trigger-free recording. Shaded band: plausible visual-cortex HbO range. The curve gives the probability that a synthetic response of a given amplitude, injected at the hypothesised onsets, would be recovered by the same pipeline applied to the real data.

4.4 Companion tool: detectability as a pre-study check

Figure 2 illustrates the calculator on two informal, self-collected trigger-free pilot sessions (16 long channels, 50 Hz, natural-image viewing, no hardware triggers recorded; see §3.6 for provenance and release decision). Injecting synthetic responses at the nominal design's onsets yields a detection floor of 0.24 μM peak HbO for Session A and 0.35 μM for Session B at 80 % power. On the real data, no stimulus-locked structure was recovered: the maximum template correlation was 0.139 and 0.140, against phase-randomised null means of 0.145 and 0.139 ($p = 0.58$ and 0.50); a free-period search over 20–45 s gave $p = 0.14$–0.87; and three unsupervised segmenters recovered no protocol-like structure. The design could thus have detected a strong response but not a weak one, and—because the onsets were never recorded—the dataset cannot support stimulus-locked analysis regardless. We report this because it is exactly the judgement we would want a reviewer to demand of a decoding claim, and because the same tool converts a vague worry about statistical power into a number a study can be designed against.

5 Discussion

5.1 What a random encoder tells us

The central practical proposal of this paper is methodological: fNIRS foundation-model papers should report a randomly initialised encoder of the same architecture as a baseline. The control

is cheap—ten seeds cost minutes on a laptop CPU—and it changes the interpretation of a result entirely. In our data the absolute accuracy of MtReNIRS (0.469) looks like a respectable result for three-class fNIRS decoding on a sparse montage. Against a random encoder (0.448 ± 0.010) it is a 2-pp difference that 17 folds cannot resolve.

This does not mean the model is without value. It means the *claim* that pretraining is responsible for the performance is not supported by this experiment. Distinguishing the two requires either much more statistical power, or datasets on which transfer is larger. Neither is available to us here, and we suspect neither is available in the current fNIRS literature.

The finding also explains why the paper's own reported gains might be small and heterogeneous: the model authors report cross-subject improvements of roughly +5.0, +4.3 and +1.2 pp on different datasets, none individually significant. Our independent estimate on a clean dataset is in the same range, which is at least consistent.

5.2 Architecture, not pretraining

The significant contrast in our results is the transformer representation against hand-crafted features, and it survives with a random encoder. This is the random-features phenomenon [9] reappearing in a new modality: a sufficiently expressive fixed representation can be competitive with a learned one when the downstream task is small. For practice, this suggests that a well-designed encoder architecture may deliver most of the value that foundation-model framing promises, at a fraction of the data-curation cost—and that the field's claims should be framed accordingly.

5.3 Systemic signal and the cost of cleaning

Our short-separation ablation is the result we least expected. Physiological regression is recommended, is well motivated, and helps elsewhere [7]. Here it reduced accuracy, and—more tellingly—it removed precisely the part of the signal on which the embedding arms were outperforming classical features. The most plausible account is that the auditory conditions differ in global physiology (arousal, attention, global haemodynamics) in addition to local cortical responses, and short channels capture that global component.

This has two implications. First, studies that use short-separation regression should report decoding performance *with and without* it; a large drop is informative about where the decodable variance lives. Second, and more broadly, it reinforces the case for interpreting fNIRS decoding results alongside explicit tests of non-neural confounds.

5.4 Integration must be learned, not concatenated

Our stacking result is a negative one, and we report it because the temptation to combine representations is strong. At matched dimensionality, concatenation degraded performance and was statistically indistinguishable from the weaker component alone. The redundancy between hand-crafted features and transformer embeddings is high; adding a representation is not the same as adding information. A combined model would need a learned routing mechanism, and it should be evaluated against the same random-encoder control.

5.5 Implications for evaluation practice

We suggest three concrete practices for fNIRS foundation-model papers: (i) a random-encoder baseline with multiple seeds; (ii) explicit contamination accounting—which evaluation datasets overlap the pretraining corpus; (iii) reporting of preprocessing sensitivity, in particular short-separation

regression, because it can change both absolute accuracy and the ranking of methods. Each is inexpensive and each materially changes what a reported number means.

6 Limitations and threats to validity

Statistical power. Seventeen folds give limited power. The 95 % confidence interval on the central contrast (pretrained versus random-init) spans roughly ±3 pp, so an advantage smaller than that cannot be resolved. Our result is a failure to establish an advantage, not evidence of its absence. Increasing folds requires more participants, not more seeds.

One dataset, one task, one model. We audit MtReNIRS on a three-class auditory block-design dataset. We did not evaluate GeoNIRS, nor a visual or affective paradigm. The auditory paradigm may be unusually favourable to systemic signal, which would make the short-separation result paradigm-specific. Generalisation across datasets and tasks is the obvious next experiment and we do not claim it here.

The random-encoder control is a lower bound. Our control re-initialises each module with its own default initialisation and runs ten seeds. A stronger control would tune the random encoder's readout as carefully as the pretrained one, which could raise it further; a weaker control (single seed) could understate it. We report the seed distribution so the reader can judge.

Probe hyperparameters are fixed. All arms use $C = 1$, PCA-64 and one solver setting. We did not perform nested hyperparameter selection, because doing so identically for twelve arms was not affordable in our compute budget. A tuned probe could change the ranking between arms whose accuracies differ by a few points. Total dimensionality is matched, which removes the most obvious artefact but not this one.

Preprocessing choices are defaults. The modified Beer–Lambert constants, the differential pathlength factor of 6.0, the 0.01–0.2 Hz band and the run-mean intensity reference are defensible defaults, not optimised values. Because epochs are z-scored per participant and channel, the short-separation ablation acts on waveform shape rather than absolute amplitude; a pipeline that preserved amplitude might show a different pattern.

Contamination accounting is by published list. We determine that the evaluation dataset is absent from the pretraining corpus from the twelve dataset names published with the model. If the dataset entered pretraining under an unrecognised name, the pretrained arm would be advantaged, so this uncertainty works in the model's favour, not against it.

Inference-only audit. We evaluate frozen embeddings and a linear probe; we do not fine-tune the encoder. Fine-tuning could change the picture substantially and is a natural follow-up. We also do not claim any result about the model's published benchmark numbers, which we did not attempt to reproduce exactly, because our preprocessing and probe differ.

Compute and numerical detail. All inference and probes ran on a laptop CPU. One earlier run of the stacking ablation used a probe that produced convergence warnings; we re-ran it under the final protocol and report only the re-run (the qualitative conclusion was unchanged). One earlier

run of the short-separation variant silently failed to change the data because the short channels had been removed before regression; the current pipeline asserts that the cleaning step changes the input, and we verified the effect numerically (correlation $0.434 \rightarrow 0.000$). We mention both because they are ordinary failure modes of this kind of pipeline and are worth guarding against.

7 Conclusion

We audited a released fNIRS foundation model on a clean held-out dataset. The pretrained encoder reached 0.469 ± 0.095 three-class accuracy against chance 0.333, but a randomly initialised encoder of the same architecture reached 0.448 ± 0.010 over ten seeds—a 2.0 pp difference that is not statistically significant. The architecture itself did significantly outperform hand-crafted features, and that advantage disappeared once short-separation channels were regressed out, at a cost of 4.5 pp. Naive concatenation of representations significantly degraded performance.

The methodological conclusion is modest and, we think, portable: fNIRS foundation-model claims should be reported against a random-encoder baseline, with contamination accounting and with preprocessing sensitivity. The empirical conclusion is deliberately narrow: on this dataset and task, the pretraining advantage of a fNIRS foundation model is not established.

Three extensions follow naturally. First, replicate across datasets and paradigms, including a visual or affective domain the model's authors never tested. Second, fine-tune rather than probe, and evaluate under the same controls. Third, replace naive concatenation with learned fusion, and test whether it beats the better single representation. We release the harness, the frozen protocol and a detectability calculator so that all three can be attempted without rebuilding the infrastructure.

Data and code availability

All analysis code, the frozen evaluation protocol, per-fold results and figure-generation scripts are available at `https://github.com/curiousbrutus/fnirs-foundation-model-audit` and archived on Zenodo (DOI to be added on acceptance). The auditory dataset analysed in §4 is publicly available from its original authors [8].

The trigger-free pilot recordings used only to illustrate the detectability calculator (§4.4) are *not* released. They are informal, self-collected exploratory recordings made by the author, obtained without a formal ethics protocol or research consent, and they are reported exclusively as aggregate statistics under neutral session codes.

Ethics statement

This study re-analyses previously collected, publicly available, de-identified data; no new human participants were involved and no new ethical approval was required.

Author contributions

E.G. conceived the study, performed the analysis and wrote the manuscript. `[TODO: add co-authors and CRediT roles before submission.]`

Use of AI tools

Generative AI tools were used in the preparation of this work: a coding assistant was used to implement, debug and test the analysis pipeline, and a language model was used to draft and edit prose. The research question, the pre-registered evaluation protocol, the statistical choices and all interpretations are the author's. Every number reported here was recomputed by the author from the released code and result files, and the author takes full responsibility for the content of the manuscript.

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